Rational Exponents — Practice

Convert · Simplify · Apply

§1 Definition of Rational Exponent: $a^{1/n}$

Where does $a^{1/n}$ come from? Start from something we know:

$$2^{10} = 1024 \quad \Rightarrow \quad (2^{10})^{\frac{1}{10}} = 1024^{\frac{1}{10}} \quad \Rightarrow \quad 2^{10\cdot\frac{1}{10}} = 1024^{\frac{1}{10}} \quad \Rightarrow \quad 2 = 1024^{\frac{1}{10}}$$

But we already know $2 = \sqrt[10]{1024}$. So:

$$a^{1/n} = \sqrt[n]{a} \qquad (n\text{th root of } a)$$

Examples: $9^{1/2} = \sqrt{9} = 3$ $\quad 8^{1/3} = \sqrt[3]{8} = 2$ $\quad 16^{1/4} = \sqrt[4]{16} = 2$ $\quad 32^{1/5} = \sqrt[5]{32} = 2$

§2 General Rule: $a^{m/n}$

When the exponent is a fraction $\frac{m}{n}$ (not just $\frac{1}{n}$), **two equivalent paths**:

$$a^{m/n}$$

$$= \big(\underbrace{a^{1/n}}_{\text{root first}}\big)^m = \big(\sqrt[n]{a}\big)^m \quad \textit{(same result)} \quad = \big(\underbrace{a^m}_{\text{power first}}\big)^{1/n} = \sqrt[n]{a^m}$$

Tip: Root first keeps numbers small! $27^{4/3}$: root first $\Rightarrow (\sqrt[3]{27})^4 = 3^4 = 81$ vs power first $\Rightarrow \sqrt[3]{27^4} = \sqrt[3]{531441} = 81$ ✓

§3 Improper Fraction Exponent — Split the Whole Part

When $m > n$ (improper fraction exponent), split off the whole number part:

$$d^{5/3} = d^{1+2/3} = d^1 \cdot d^{2/3} = d \cdot \sqrt[3]{d^2}$$

General strategy:

$$a^{p/q} = a^{\lfloor p/q \rfloor + r/q} = a^{\lfloor p/q \rfloor} \cdot a^{r/q} \qquad \text{where } p = \lfloor p/q \rfloor \cdot q + r$$

Another example: $x^{7/4} = x^{1+3/4} = x \cdot x^{3/4} = x \cdot \sqrt[4]{x^3}$

▷ Caution — Parentheses Matter!

WITHOUT parentheses:

$$5x^{1/3} = 5 \cdot \sqrt[3]{x}$$

Only x is raised to the power $\frac{1}{3}$.

WITH parentheses:

$$(5x)^{1/3} = \sqrt[3]{5x}$$

Both 5 and x are raised to the power $\frac{1}{3}$.

Memory trick: the exponent only "reaches" what is immediately to its left.

Big Ideas: $a^{1/n} = \sqrt[n]{a}$ $a^{m/n} = (\sqrt[n]{a})^m = \sqrt[n]{a^m}$ **Improper: split the whole part first**

Group 1 Converting Between Forms

1. Evaluate directly: $(1024)^{1/10} = \sqrt[10]{1024} =$ _____

2. Write $a^{5/6}$ in radical form: _____

3. Verify both forms give the same answer for $a^{5/6}$: $\left(\sqrt[6]{a}\right)^5 = \sqrt[6]{a^{\square}} =$ _____

4. Rewrite with a rational exponent: $\sqrt[3]{5x} =$ _____

5. Rewrite with a rational exponent: $5 \cdot \sqrt[3]{x} =$ _____ *Note: different from #4!*

6. Convert to radical form: $x^{2/5} =$ _____

7. Convert to exponential form: $\sqrt[4]{x^3} =$ _____

Group 2 Evaluate / Simplify (Numbers)

8. $8^{2/3} = \left(\sqrt[3]{8}\right)^2 =$ _____

9. $27^{4/3} = \left(\sqrt[3]{27}\right)^4 =$ _____

10. $16^{3/4} = \left(\sqrt[4]{16}\right)^3 =$ _____

11. $64^{2/3} = \left(\sqrt[3]{64}\right)^2 =$ _____

12. $81^{3/4} = \left(\sqrt[4]{81}\right)^3 =$ _____

Group 3 Simplify Algebraic Expressions

13. $d^{5/3} = d^{1+2/3} = d \cdot d^{2/3} = d \cdot$ _____

14. $\left(x^{1/2}\right)^6 = x^{1/2 \times 6} =$ _____

15. Rewrite with a single exponent: $\sqrt[3]{\sqrt{x}} = \left(x^{1/2}\right)^{1/3} = x^{1/\square} =$ _____

16. Simplify and write in radical form: $\dfrac{x^{3/4}}{x^{1/4}} = x^{3/4\,-\,1/4} = x^{\square} =$ _____

17. Simplify: $\left(\dfrac{a^3}{a^5}\right)^4 = (a^{3-5})^4 = (a^{-2})^4 =$ _____

Group 4 Prime Factorisation & Multi-step

18. Simplify $(54y)^{2/3}$:

$$(54y)^{2/3} = _____$$

19. Expand and simplify $(54y^2)^{2/3}$. *Hint:* $54 = 2 \cdot 27 = 2 \cdot 3^3$

$$(54y^2)^{2/3} = (2 \cdot 3^3)^{2/3} \cdot y^{2/3 \cdot 2} = 2^{2/3} \cdot 3^{3 \cdot 2/3} \cdot y^{4/3} = _____$$

Group 5 True / False & Challenge

20. **True or False** — explain your reasoning:

$$4x^{1/2} \stackrel{?}{=} (4x)^{1/2}$$

Answer: _____

Reason: _____

21. **Challenge** — simplify completely *(apply exponent to numerator and denominator separately)*:

$$\left(\frac{8x^3}{27y^6}\right)^{2/3} = \frac{(8x^3)^{2/3}}{(27y^6)^{2/3}} = \frac{8^{2/3} \cdot x^{3 \cdot 2/3}}{27^{2/3} \cdot y^{6 \cdot 2/3}} = _____$$

Answer Key — Teacher Copy

Group 1 — Converting

1. **2** $(2^{10} = 1024 \Rightarrow 1024^{1/10} = 2)$
2. $\sqrt[6]{a^5}$
3. $\sqrt[6]{a^5}$ ✓ both forms match
4. $(5x)^{1/3}$ (both 5 and x inside)
5. $5x^{1/3}$ (only x raised)
6. $\sqrt[5]{x^2}$
7. $x^{3/4}$

Group 2 — Evaluate

8. **4** $(\sqrt[3]{8})^2 = 2^2$
9. **81** $(\sqrt[3]{27})^4 = 3^4$
10. **8** $(\sqrt[4]{16})^3 = 2^3$
11. **16** $(\sqrt[3]{64})^2 = 4^2$
12. **27** $(\sqrt[4]{81})^3 = 3^3$

Group 3 — Algebraic

13. $d \cdot \sqrt[3]{d^2}$
14. x^3 $\frac{1}{2} \times 6 = 3$
15. $x^{1/6}$ $(x^{1/2})^{1/3} = x^{1/6}$
16. $x^{1/2} = \sqrt{x}$ $\frac{3}{4} - \frac{1}{4} = \frac{2}{4} = \frac{1}{2}$
17. a^{-8} $(a^{-2})^4 = a^{-8}$

Group 4 — Prime Factorisation

18. $\sqrt[3]{(54y)^2} = \sqrt[3]{2916\,y^2}$
19. $2^{2/3} \cdot 3^2 \cdot y^{4/3} = 9 \cdot \sqrt[3]{4y^4} = 9\sqrt[3]{4} \cdot y\sqrt[3]{y}$ $\left(y^{4/3} = y \cdot y^{1/3} = y\sqrt[3]{y}\right)$

Group 5 — True/False & Challenge

20. **False.** $4x^{1/2} = 4\sqrt{x}$ but $(4x)^{1/2} = \sqrt{4x} = 2\sqrt{x}$. They differ by a factor of 2.

21. $\dfrac{8^{2/3} \cdot x^2}{27^{2/3} \cdot y^4} = \dfrac{(\sqrt[3]{8})^2 \cdot x^2}{(\sqrt[3]{27})^2 \cdot y^4} = \dfrac{4x^2}{9y^4} = \boxed{\dfrac{4x^2}{9y^4}}$